BioNet | Demonstration, Viva & Submission Pack

Three-minute presentation script

0:00–0:30 — Introduction

“Our project BioNet implements reliable file transmission over UDP using Selective Repeat ARQ. The assignment requires transmission of actual captured biometric data and exact reconstruction. UDP does not guarantee delivery or order, so we add acknowledgements, per-packet timers, retransmissions and receiver buffering.”

0:30–1:05 — Architecture

“The sender reads a ZIP file, computes SHA-256, divides it into 1,024-byte chunks and transmits them using a sliding window. Every packet has a sequence number and CRC32. The receiver validates packets, acknowledges them individually, buffers out-of-order arrivals and writes data in the original order.”

1:05–1:40 — Live synthetic demonstration

“We run a receiver and sender over actual UDP sockets. We simulate packet loss. The dashboard shows progress, acknowledged packets, retransmissions and the window. When packet 1 is lost but 2 and 3 arrive, the receiver buffers the later packets; only missing packet 1 must be retried.”

1:40–2:10 — Actual biometric evidence

“On a MacBook Air M4 we transferred our real 2,774,235,808-byte biometric ZIP using 5% simulated outbound loss. All 2,709,215 DATA packets were acknowledged, with 143,000 retransmissions. Both endpoints reported SHA-256 PASS, and independent cmp returned byte-for-byte PASS.”

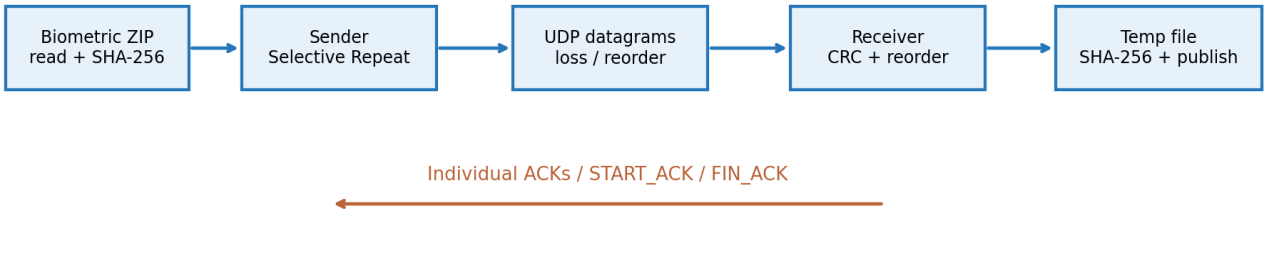
2:10–2:40 — Performance

“Our 90 synthetic Phase 3 benchmarks all completed with verified hashes. In the controlled 64 KiB window-32 sweep, mean goodput fell from 22.596 Mbps at 0% simulated loss to 0.252 Mbps at 40%. TCP was faster than our Python UDP implementation on unimpaired localhost; our goal is demonstrating reliable selective retransmission, not UDP superiority.”

2:40–3:00 — Conclusion

“BioNet demonstrates reliable reconstruction with packet loss, reordering and integrity checks. Our 181 macOS tests passed. Limits include finite retries, no encryption or authentication, and a localhost real-data demonstration rather than a physical LAN transfer.”

Architecture & Selective Repeat example



Actual sockets on localhost for verified biometric transfer; LAN test not yet demonstrated

Illustrative sequence (not a measured packet trace)

Sender	Network / Receiver	Effect
DATA 0	Received; ACK 0	Base moves to 1
DATA 1	Simulated drop	Hole at sequence 1
DATA 2, DATA 3	Received; ACK 2, ACK 3	Receiver buffers later packets
Timeout DATA 1	Retransmit DATA 1; ACK 1	Buffered 2,3 become contiguous
FIN	Receiver checks size and SHA-256; FIN_ACK	Verified output published

Common viva questions (short answers)

Why UDP? It exposes unreliable datagrams so we can implement and demonstrate ARQ ourselves.

Why Selective Repeat instead of Go-Back-N? Only missing/unacknowledged packets are retransmitted, not all packets after a loss.

What does the sliding window do? It allows several packets to be outstanding, improving pipeline utilization.

Why individual ACKs? They identify which particular packets arrived successfully.

What happens when ACK is lost? The sender retries on timeout; receiver recognizes duplicate DATA and ACKs again.

Why CRC32 and SHA-256? CRC32 checks individual datagrams; SHA-256 checks the reconstructed whole file. Neither authenticates the sender.

How is out-of-order delivery handled? A bounded reorder buffer holds later packets until the missing earlier chunk arrives.

How is goodput calculated? Useful file bytes × 8 divided by transfer seconds and 1,000,000.

Why was the real transfer slow? The Python implementation, fixed timeout, small chunks, window size and simulated loss affect completion time; no causal breakdown was measured.

Was the real test over a LAN? No. It used real UDP sockets over localhost on one Mac.

Is biometric data secure on the network? Not against eavesdropping: the application lacks encryption and authentication; use only a trusted environment.

What did you verify independently? Both endpoints reported SHA-256 PASS and macOS cmp confirmed byte-for-byte equality.

Submission checklist & screenshots

Before submission, capture the following from your actual Mac; no screenshot is fabricated in this pack.

- ■ Sender dashboard showing 100% and SHA-256 PASS (crop out unrelated private windows).
- ■ Receiver JSON showing SHA-256 PASS and bytes_written.
- ■ Terminal line: BYTE-FOR-BYTE VERIFICATION: PASS.
- ■ Terminal line: 181 passed in 30.79s.
- ■ At least two measured plots from the attached charts.
- ■ Architecture diagram and two-page technical report.
- ■ Source code / requirements / experiment conditions and CSV evidence.
- ■ Instructor approval for Selective Repeat, if not yet obtained.
- ■ If required by the evaluator: separate two-device trusted-LAN demonstration using synthetic data first.

Useful local commands

```
cd "$HOME/vscode/CN Project/BioNet"
source .venv/bin/activate
PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 python -m pytest -q
python -m scripts.demo --output results/final-demo
```

Keep the real ZIP and received copy off GitHub and out of submission archives unless the instructor explicitly requires a secure handoff. The real transfer logs are user-supplied observations; the synthetic benchmark figures come from the repository's Phase 3 recorded results.